Inverted MAML Card 1: Round 12 Thermal Bit-Slicing Report

Hardware: Card 1 (T120+T20, 1k/2.2k) | **FW:** v2.2-MAML (0x20261030) | **Params:** 40% PWM, K=1, LR=0.055, L2=1920, Mom=0.85, Clamp=±80 LSB

1. Round 12 Summary & Cumulative Comparison (R1–R11)

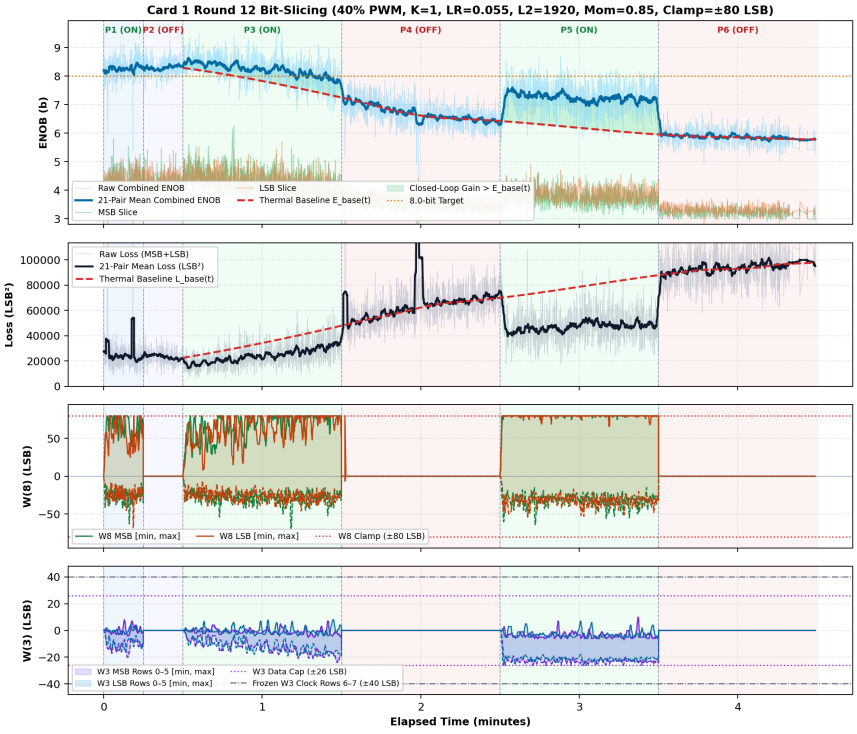
• **Cold Verification:** Phase 1 ON = **9.45 b**, Phase 2 OFF = **9.45 b** (W8=0 strictly).

• **Round 12 Detrended Trend Metrics:** Lift vs E_base = **+0.95 b**, Loss Ratio = **1.70x**, Climb = **+0.98 b**, Jitter = **0.42 b**, Recovery = **0.00**, Score = **+1.810**.

Rnd	LR	L2	Mom	Clamp	Lift	LossR	Climb	Jitt	Recov	Score
R4dc_fix	0.05	1792	0.85	±70	+1.08b	1.78x	+0.97b	0.40b	0.50	+2.480
R5dc_fix	0.055	1792	0.85	±85	+1.12b	1.77x	+1.37b	0.42b	0.00	+2.393
R6dc_fix	0.055	1792	0.8	±90	+0.99b	1.72x	+1.02b	0.40b	0.00	+1.770
R7dc_fix*	0.055	2048	0.85	±75	+1.04b	1.74x	+1.17b	0.40b	0.50	+2.665
R8dc_fix	0.055	2048	0.85	±75	+1.16b	1.85x	+0.68b	0.42b	0.00	+1.855
R9dc_fix	0.055	1920	0.85	±75	+0.81b	1.59x	+0.66b	0.38b	0.50	+1.756
R10dc_fix	0.055	1920	0.85	±78	+0.96b	1.67x	+1.13b	0.40b	0.00	+1.898
R11dc_fix	0.055	1920	0.85	±80	+0.95b	1.70x	+0.98b	0.42b	0.00	+1.810

2. Phase-by-Phase Telemetry (3,941 Pairs)

Phase	Regime	N	Start->Late (Mean)	Δ vs E_base	Loss Start->Late	W8 Pk [M,L]	W3 Pk	W8 nz
P1	Cold / Regler ON	319	8.10->8.32 (8.26b)	-	27839->27117	[80,80]	20	319
P2	Cold / Regler OFF	278	8.18->8.35 (8.29b)	-	27416->21779	[0,0]	0	0
P3	Warm / Regler ON	1026	8.50->8.02 (8.24b)	+0.54 b	18941->29068	[80,80]	24	1026
P4	Warm / Regler OFF	874	7.22->6.48 (6.72b)	-0.00 b	48112->68745	[0,80]	0	1
P5	Warm / Regler ON	822	6.54->7.20 (7.23b)	+1.17 b	67677->48345	[80,80]	26	822
P6	Warm / Regler OFF	622	5.94->5.82 (5.87b)	+0.01 b	89005->96126	[0,0]	0	0



3. All W(8) (7x6 MSB+LSB) & W(3) (8x6 MSB+LSB) Weight Dynamics — Round 12

• **Full 180-Synapse Per-Pair Capture:** Every individual weight in **Matrix 8 MSB (42 weights)**, **Matrix 8 LSB (42 weights)**, **Matrix 3 MSB (36 active + 12 clock)**, and **Matrix 3 LSB (36 active + 12 clock)** is plotted below across all columns `Col 0..5`. W(8) enforces anti-windup clamping at ±80 LSB, while W(3) rows 0–5 stay within ±26 LSB so rows 6–7 clock staircases remain locked at ±40 LSB.

Round 12 Full Hardware Weight Telemetry: ALL 84 W(8) [7x6 MSB+LSB] & ALL 96 W(3) [8x6 MSB+LSB] Synapses

